

Hierarchical Data Structure for Genetic Diseases Using Tree Algorithms

Course: Data Structures and Algorithms
Asal Rabiee

Abstract:

This project involves creating a **hierarchical tree data structure** to represent diseases and associated genes, with attributes for each gene. Implemented in Python, it includes building nodes, trees, and hash tables to store gene details, with user-driven retrieval of information. This approach offers a clear, organized way to represent and manage complex biological data, enabling users to navigate diseases and obtain detailed genetic information. Such data structures (and similar ones) are widely used in biological databases such as **OMIM** and **NCBI**.

Introduction:

As control over infectious and nutritional diseases has improved in developed countries, it has become evident that **genetic diseases** are a major cause of disability, mortality, and human tragedy. Very few families are entirely free of genetic disorders. Thousands of genetic disorders with defined clinical symptoms have been identified, ranging from:

- **Chromosomal number abnormalities**
- **Chromosomal structural abnormalities**
- **Sex chromosome abnormalities** [1]

A genetic disorder arises from errors in DNA, which can include:

- Single nucleotide mutations
- Single-gene or multi-gene mutations
- Whole chromosome deletions or additions

These disorders are classified into:

- Monogenic disorders
- Chromosomal disorders
- Complex disorders [2]

With advances in **data structures**, interfaces can now integrate genomic data, supporting analysis and visualization. Recent efforts focus on **data content, retrieval, and visualization tools** [3]. Genome databases are essential for experimental and computational biologists, providing genotype–phenotype information from large-scale genomics projects [4]. The goal is to structure this data so gene–disease relationships can be easily explored in future scientific studies.

Project Goal:

The main objective is to design a **hierarchical data structure** to systematically represent:

- Genetic diseases
- Their subcategories
- Associated genes

This structured approach simplifies the organization of complex biological data, facilitating navigation, searching, and retrieval of disease–gene information.

Methodology:

To model diseases and genes, **tree or graph algorithms** are applied [5]. The structure allows users to begin at a disease node and navigate through subcategories down to individual genes, retrieving attributes for each gene.

Tree Structure

- **Root:** Represents all genetic diseases.
- **Intermediate nodes:** General and specific categories of diseases.
- **Leaves:** Genes involved in specific diseases.
- **Gene attributes:** Chromosomal location, sequence, length, exons, type, official name, etc.

Example:

- Root → Genetic Diseases
 - Subcategory → Neurodegenerative diseases, cancer
 - Specific diseases → Parkinson's, Multiple Sclerosis, Glioblastoma
 - Gene nodes → Store attributes in a **hash table**

Code Summary:

Tree Class

Represents the database system with attributes and methods:

- `__init__`: Initializes empty lists and dictionaries for diseases, genes, and gene info.
- `get_int_input`: Ensures integer input from user with error handling.
- `Set_Sub_Disease`: Adds diseases entered by user and stores them in a file.
- `Set_Genes`: Adds genes linked to each disease (name, chromosome location, exons, full name, type). Saves to file.
- `Get_Sub_Disease`: Retrieves list of diseases.
- `Get_Genes`: Retrieves gene dictionary.
- `write_to_file`: Saves data into a file `DS-Project.txt`.
- `Search_Genome_of_Disease`: Retrieves genes associated with a specific disease.

Menu Functionality

The menu provides 3 options:

1. **Create database** → Add diseases & genes via `Set_Sub_Disease` and `Set_Genes`.
2. **Search disease genome** → Find genes related to a disease using `Search_Genome_of_Disease`.
3. **Exit** → End the program.

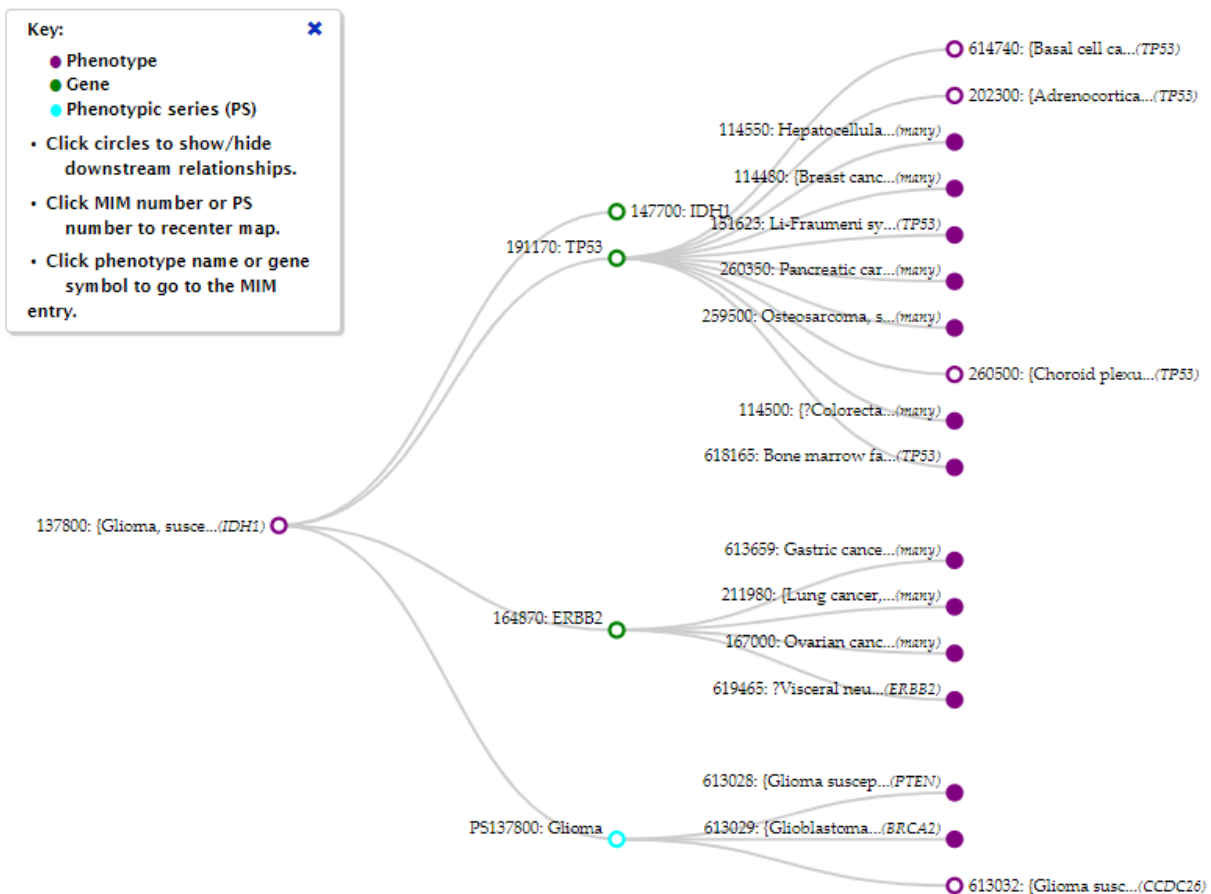
Program runs in an infinite loop until “Exit” is selected.

Example:

Disease: Glioblastoma

Key associated genes in humans:

- **ERBB2**: 35 exons, location 17q12, protein-coding, *erb-b2 receptor tyrosine kinase 2*
- **TP53**: Chromosome 17, 13 exons, location 17p13.1, protein-coding, *tumor protein p53*
- **IDH1**: Chromosome 2, 12 exons, location 2q34, protein-coding, *isocitrate dehydrogenase*



Reference: [OMIM database](#)

References

1. Fridovich, Irwin; Fridovich-Keil, Judith L.; Robinson, Arthur. (2023). *Human genetic disease*. Encyclopedia Britannica.
2. Haspolat, Yusuf; Atabay, Ayten; Aşır, Fırat. (2023). *Functional Medicine Part 5*.
3. Wegrzyn JL, Lee JM, Tearse BR, Neale DB. (2008). *TreeGenes: A forest tree genome database*. Int J Plant Genomics.
4. Wegrzyn, J.L., Main, D., Figueroa, B. et al. (2012). *Uniform standards for genome databases in forest and fruit trees*. Tree Genetics & Genomes 8, 549–557.
5. Kabari, Ledisi & Nwachukwu, Enoch. (2012). *Neural Networks and Decision Trees For Eye Diseases Diagnosis*.